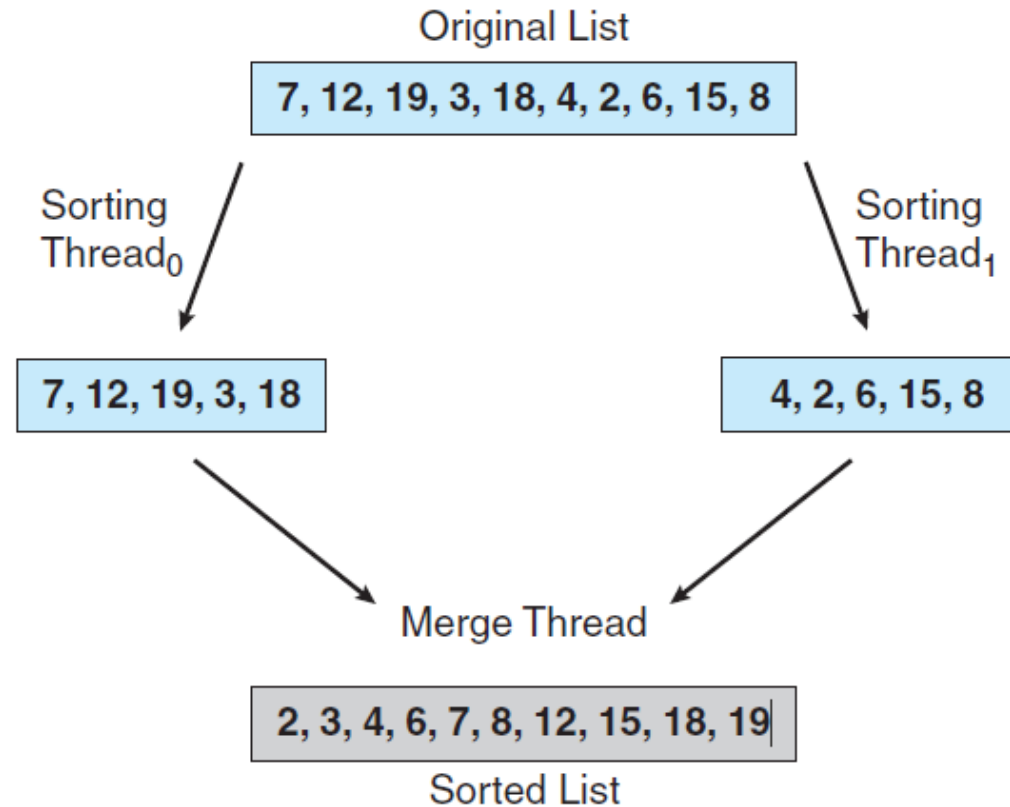


Multithread Programming

Multithreaded Sorting Application

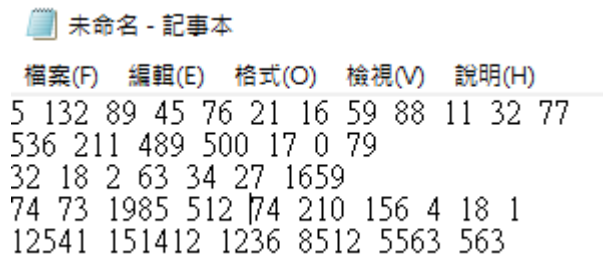
- In this project, you need to write a multithreaded sorting program that works as follow:
 - A list of integers is divided into **two smaller lists of equal size**. You have to create **two separate threads(sorting threads)** to sort each sublist using the **merge sort** algorithm.
 - Then, the two sublists are merged into a single sorted list by a **third thread(merging thread)**.

Multithreaded Sorting Application



Multithreaded Sorting Application

- We will give you a .txt file which contains several lines of **integers** as your input test data. Each line represents one test case which is composed by several numbers. So, you should figure out **how to use argc, argv way to read in the test data line by line first.**
- E.g.



未命名 - 記事本

檔案(F) 編輯(E) 格式(O) 檢視(V) 說明(H)

```
5 132 89 45 76 21 16 59 88 11 32 77
536 211 489 500 17 0 79
32 18 2 63 34 27 1659
74 73 1985 512 74 210 156 4 18 1
12541 151412 1236 8512 5563 563
```

Multithreaded Sorting Application

- Then, you need to implement the merge sort using multithreaded programming : **two threads for sorting, and one for merging**.
 - Make sure that the third thread(merging thread) get started after both sorting threads are done, which means you have to **keep the merging thread waiting until the sorting threads finish their jobs by pthread condition wait**.
 - Trace **pthread.h** first.
- Finally, you should **print out your running time for each test case on the screen, and output a .txt file for your sorting result**.
 - The output filename must be **output.txt**.
 - Please output your result with **the same format as the input** we gave you.

Multithreaded Sorting Application

- We will use more difficult test cases to test your program.
 - At most 10000 integers for one test case.
 - You can find some test cases to test your program by yourself.
- You should implement under Linux.
- Only merge sort is acceptable.
- Must use argc, argv way to input the test file.
 - Use gcc hw2.c -pthread -o hw2.o to compile your code.
 - Your command to run and output your code must be ./hw2.o testcase.txt output.txt

Multithreaded Sorting Application

- hw2_{studentID}.rar :
 - hw2.c (90%)
 - 10% for each hidden test cases, so you need to pass 9 test cases for full score 😊.
 - hw2_report (10%)
 - Tell us how you implement your homework and show us your time and result with some screenshots.

Multithreaded Sorting Application

- We will randomly pick 1/4 of all students to demo in person after the midterm.
- 0 will be given to cheaters, so don't copy & paste your friend's code directly.
- Make sure that you totally understand your code 😊.
- Delay is allowed.
 - $\text{score} = (1 - 10\% * \text{\#delay days}) * \text{original score}.$